RETAIL SALES ANALYSIS

From sales data to business decisions

Understand the data, explore the results,
and turn them into simple recommendations.

THE DECISION PATH

The analysis followed 3 simple steps

The analysis moved from understanding the data to making simple recommendations.

01

Understand

Understand the sales data and business questions.

02

Explore

Look at sales by products, categories, customers, channels, and time.

03

Recommend

Turn the results into simple business recommendations.

Three clear steps keep the analysis connected to decisions.

DATA QUALITY & CLEANING

Data was cleaned and prepared before doing the analysis.

Missing data was treated.

FIELD	MISSING	TREATMENT
Transaction Date	0%	No missing data.
Item	10%	Filled in From Category and price columns.
Price per Unit	5%	Calculated using Total Spent ÷ Quantity.
Quantity	5%	Filled using the median.
Total Spent	5%	Calculated using Price × Quantity.
Discount Applied	33%	Labeled as Unknown.

- Data type: Transaction Date Was converted to date format
- Outliers: Unusual Total Spent values were kept because they were valid, real transactions.

INTERACTIVE ANALYSIS

The Streamlit dashboard makes the analysis interactive

Use the finished analysis to move from a headline KPI into the evidence behind it.

Scan

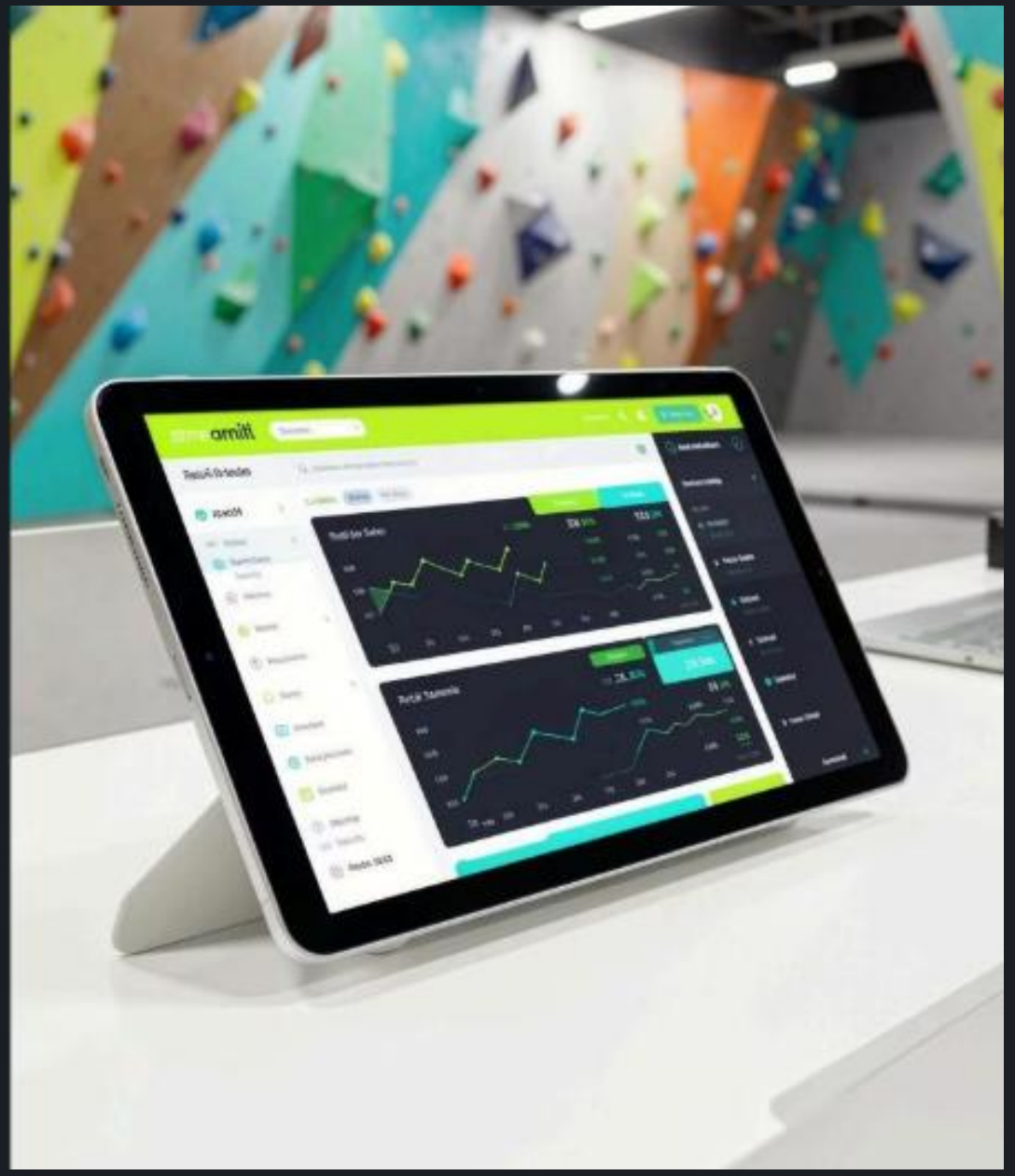
KPI cards and revenue trends

Compare

Categories, products, and discounts

Drill in

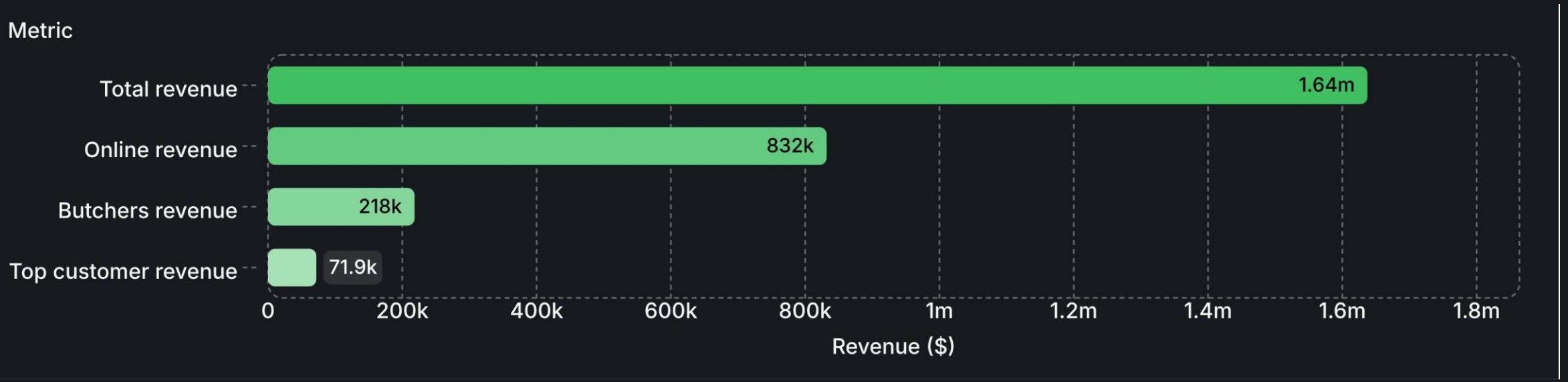
Customers, payment usage, and location split



HEADLINE PERFORMANCE

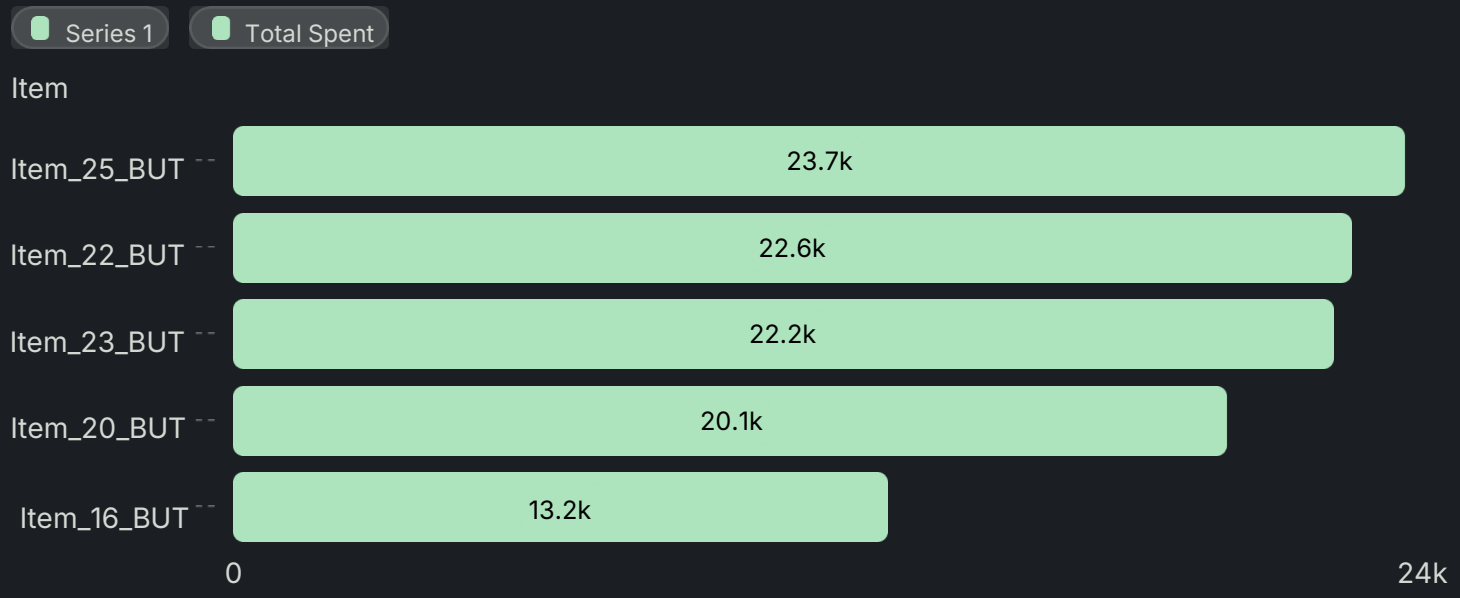
Revenue is concentrated around a few clear winners

TOTAL REVENUE	TOP CATEGORY	TOP CHANNEL	TOP CUSTOMER
$1,637,367	$218,153	$831,922	$71,890
	Butchers	Online	CUST_24



Top 5 Butchers Items by Revenue

Butchers was the best category



TOP CATEGORY

$218,153

Category revenue

BUSINESS FOCUS

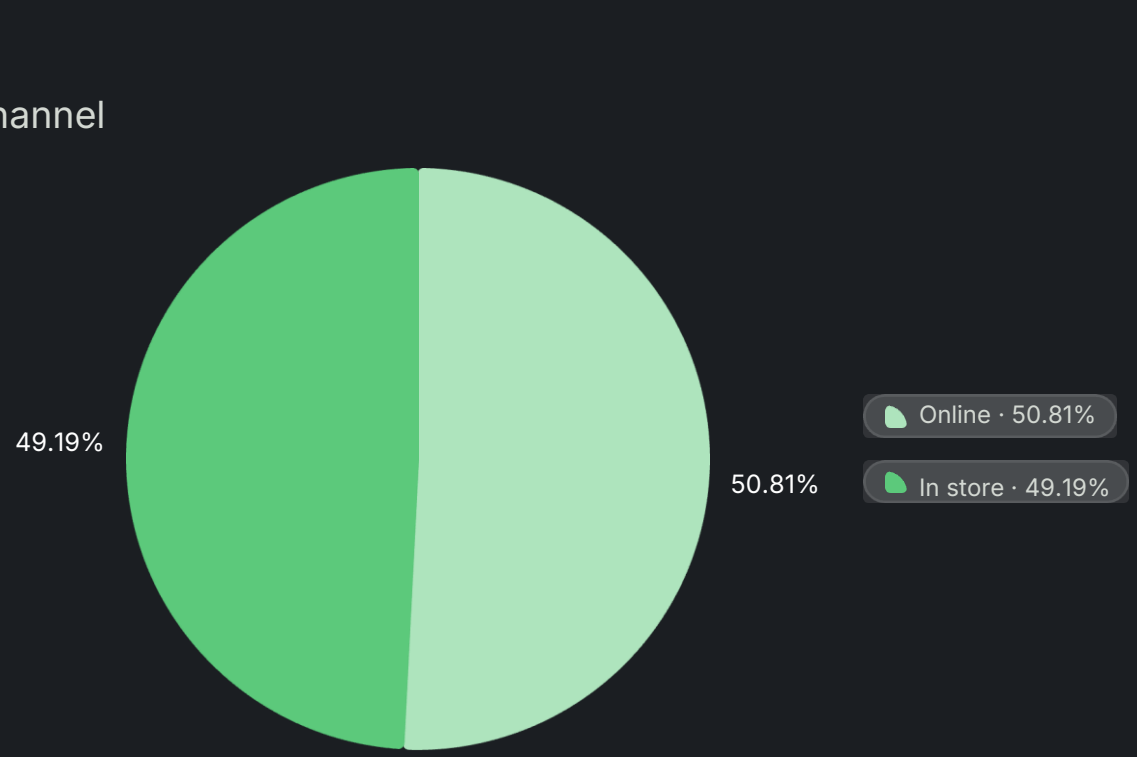
Keep it available

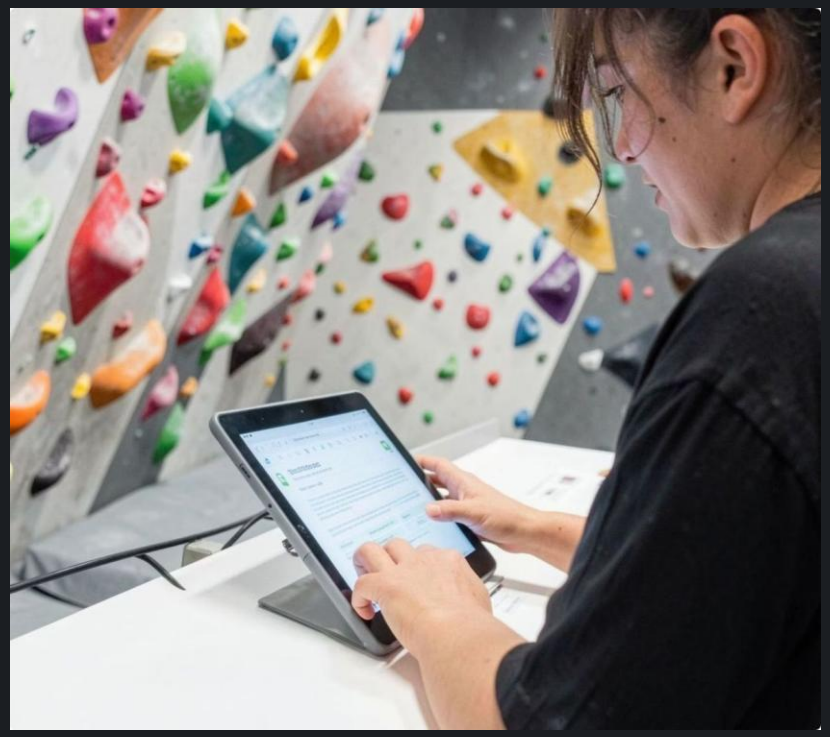
Use it as an example for other categories.



SALES CHANNEL

Online is the most important sales channel





Online revenue: **$831,922**.
Online and in store is nearly equal but online has slight edge.

To maintain this lead, make shopping easier, faster, and more reliable. Use bundles and special offers to encourage customers to buy more.

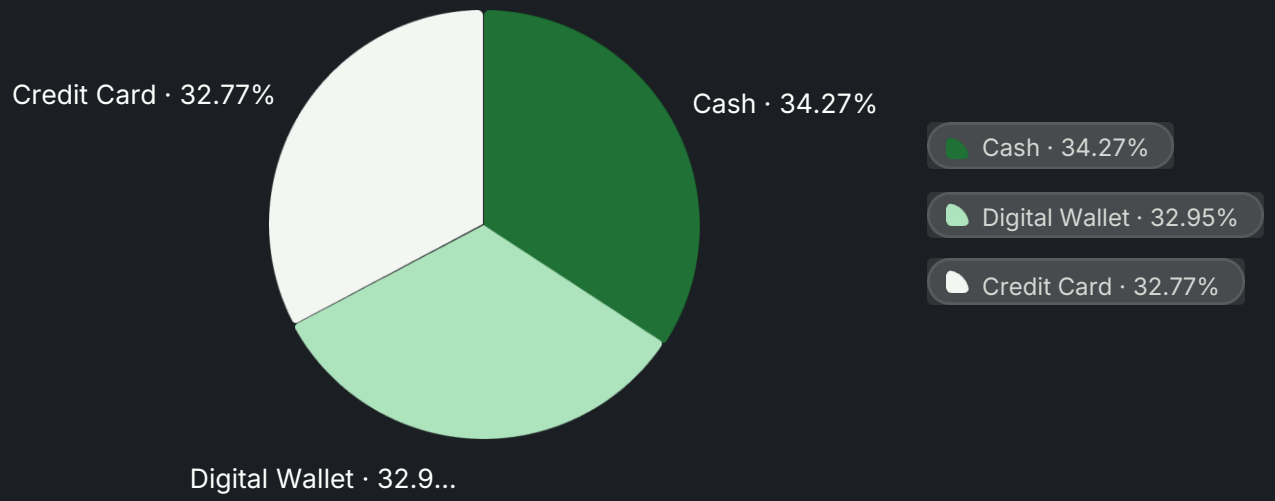
PAYMENT BEHAVIOR

Cash Leads slightly in payment method

4,310

Cash transactions

- payment methods are the most evenly split, with cash leading slightly at 4,310 transactions
- Keep cash payments
easy and encourage digital payments.



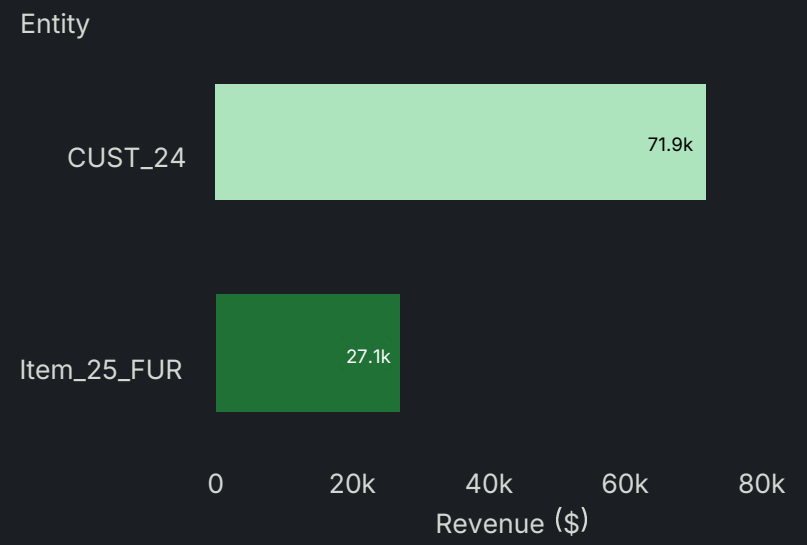
CUSTOMERS AND PRODUCTS

Some customers and products bring much more money than others

CUST_24: **$71,890** — the most important customer to keep.

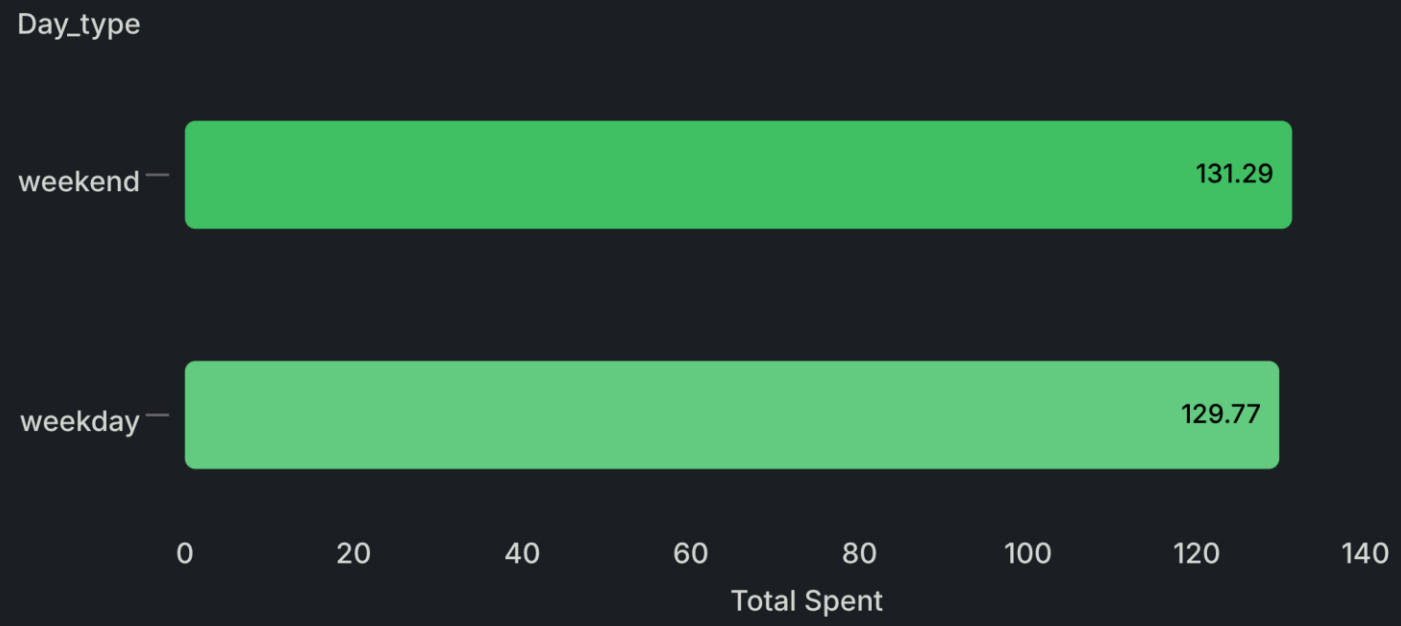
Item_25_FUR: **$27,101** — the highest-revenue product.

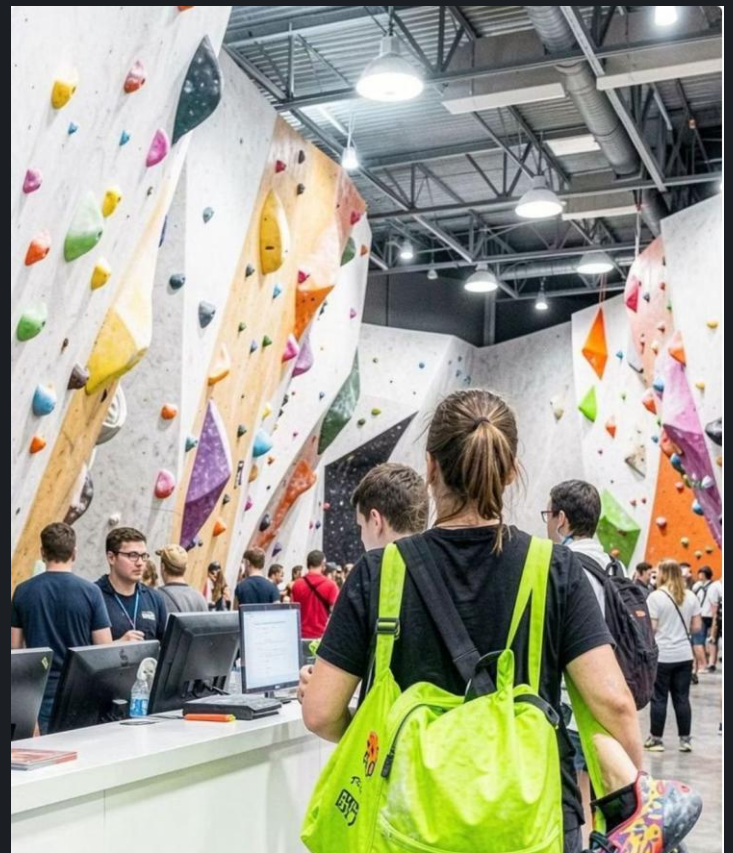
Keep an eye on these customers and products.



PURCHASING BEHAVIOR

Average Spending is Stable Across the week





Use weekends to offer product bundles, related products, and special offers